Tugas : Laporan Praktikum Mandiri 7

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```python
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

Pada bagian ini kita mengimpor semua library yang dibutuhkan dan memuat dataset

```python
df = pd.read_csv(path + "/data/dataset_satelit.csv", sep=",")
df.head()
```

	No	Longitude	Lattitude	N	P	K	Ca	Mg	Fe	Mn	...	b1	Sigma_VV	Sigma_VH	plia	lia	iafe	gamma0_vv	gamma0_vh	beta0_vv	beta0_vh
0	1	103.036658	-0.604417	2.64	0.15	0.415	0.51	0.31	119.96	463.23	...	0.0433	0.18183	0.04461	35.74446	35.79744	35.41161	0.22331	0.05479	0.31325	0.07686
1	2	103.037201	-0.604689	2.75	0.17	0.568	0.76	0.58	102.63	493.81	...	0.0465	0.22079	0.04640	35.12096	35.14591	35.41510	0.27116	0.05699	0.38033	0.07993
2	3	103.036359	-0.603012	1.77	0.12	0.339	0.49	0.6	107.37	460.93	...	0.0417	0.18926	0.03992	35.07724	35.07730	35.41135	0.23242	0.04902	0.32604	0.06876
3	4	103.036950	-0.603219	2.30	0.15	0.460	0.74	0.67	96.02	338.17	...	0.0367	0.14769	0.03622	36.08078	36.08469	35.41583	0.18138	0.04448	0.25440	0.06238
4	5	103.036802	-0.601969	2.05	0.14	0.308	0.64	0.72	87.01	384.33	...	0.0361	0.18205	0.03797	32.68855	32.69293	35.41592	0.22359	0.04664	0.31359	0.06541

5 rows × 34 columns

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 594 entries, 0 to 593
Data columns (total 34 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   No         594 non-null    int64
 1   Longitude  594 non-null    float64
 2   Lattitude  594 non-null    float64
 3   N          594 non-null    float64
 4   P          594 non-null    float64
 5   K          593 non-null    float64
 6   Ca         594 non-null    float64
 7   Mg         594 non-null    object
 8   Fe         594 non-null    float64
 9   Mn         594 non-null    float64
 10  Cu         594 non-null    float64
 11  Zn         594 non-null    float64
 12  B          594 non-null    float64
 13  b12        594 non-null    float64
 14  b11        594 non-null    float64
 15  b9         594 non-null    float64
 16  b8a        594 non-null    float64
 17  b8         594 non-null    float64
 18  b7         594 non-null    float64
 19  b6         594 non-null    float64
 20  b5         594 non-null    float64
 21  b4         594 non-null    float64
 22  b3         594 non-null    float64
 23  b2         594 non-null    float64
 24  b1         594 non-null    float64
 25  Sigma_VV   594 non-null    float64
```

-mencetak lima data pertama dan melihat info tiap data

```
df.describe()
```

	No	Longitude	Lattitude	N	P	K	Ca	Fe	Mn	Cu	...	b1	Sigma_VV	Sigma_VH	plia	lia	iafe	gamma0_vv	gamma0_vh	beta0_vv	beta0_vh
count	594.000000	594.000000	594.000000	594.000000	594.000000	593.000000	594.000000	594.000000	594.000000	594.000000	...	594.000000	594.000000	594.000000	594.000000	594.000000	594.000000	594.000000	594.000000	594.000000	594.000000
mean	297.500000	106.878644	-1.024933	2.259091	0.141380	0.582175	0.595094	74.613771	308.034697	2.391195	...	0.177291	0.234474	0.102789	28.640422	28.664891	28.609569	0.202587	0.051524	0.269642	0.062320
std	171.617307	4.949840	0.965349	0.395499	0.019782	0.222567	0.366118	55.579655	241.731643	1.580296	...	0.155615	0.070516	0.112310	15.325347	15.380384	15.329170	0.104357	0.012959	0.143728	0.024218
min	1.000000	102.760857	-2.333750	1.140000	0.090000	0.122000	0.050000	21.080000	3.160000	0.090000	...	0.014100	0.115170	0.021460	0.127000	0.098600	0.026000	0.008700	0.016900	0.009300	0.016300
25%	149.250000	102.927811	-2.233338	1.982500	0.130000	0.429000	0.320000	40.705000	124.015000	1.172500	...	0.046925	0.183210	0.039535	31.959745	31.968948	33.685353	0.183085	0.040250	0.244935	0.052772
50%	297.500000	103.581969	-0.602276	2.280000	0.140000	0.549000	0.540000	65.650000	239.445000	2.225000	...	0.072700	0.213385	0.046550	35.067930	35.110415	34.611565	0.233590	0.050415	0.310380	0.068380
75%	445.750000	113.403797	-0.257349	2.570000	0.150000	0.710000	0.790000	87.372500	434.990000	3.357500	...	0.318900	0.262242	0.059190	38.319135	38.441065	39.002760	0.271790	0.060410	0.364505	0.079020
max	594.000000	113.434700	0.069251	3.230000	0.220000	1.489000	2.820000	559.100000	2009.320000	8.170000	...	0.751400	0.512210	0.373000	47.592900	48.014640	39.209330	0.658960	0.122300	0.814170	0.150620

8 rows × 33 columns

-menampilkan ringkasan statistik kolom numerik

```
X = df[['gamma0_vv', 'gamma0_vh']]
y = df['Fe']

X.head(), y.head()
```

```
(   gamma0_vv  gamma0_vh
 0    0.22331    0.05479
 1    0.27116    0.05699
 2    0.23242    0.04902
 3    0.18138    0.04448
 4    0.22359    0.04664,
 0    119.96
 1    102.63
 2    107.37
 3     96.02
 4     87.01
 Name: Fe, dtype: float64)
```

Di sini kita memilih kolom `gamma0_vv` dan `gamma0_vh` sebagai variabel input, serta `Fe` sebagai variabel target.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
len(X_train), len(X_test)
```

```
(475, 119)
```

Data dibagi menjadi 80% untuk pelatihan dan 20% untuk pengujian.

```python
model = LinearRegression()
model.fit(X_train, y_train)

print("Intercept  :", model.intercept_)
print("Koefisien  :", model.coef_)
```

```
Intercept  : 126.849412571108
Koefisien  : [   80.20446277 -1365.46992884]
```

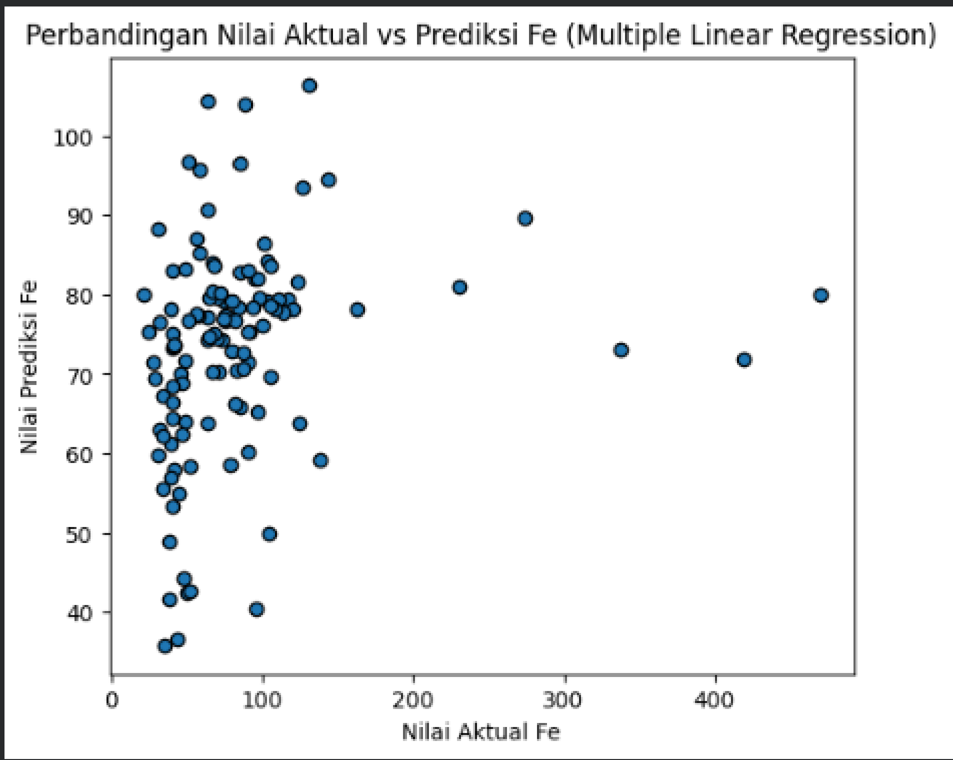
Model dilatih menggunakan data training.

```python
y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
accuracy = r2 * 100

print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R² Score:", r2)
print("Akurasi (persentase):", accuracy, "%")
```

```
MAE : 33.47415762965465
MSE : 4134.322719701519
RMSE: 64.29869920691647
R² Score: 0.02630761725809183
Akurasi (persentase): 2.630761725809183 %
```

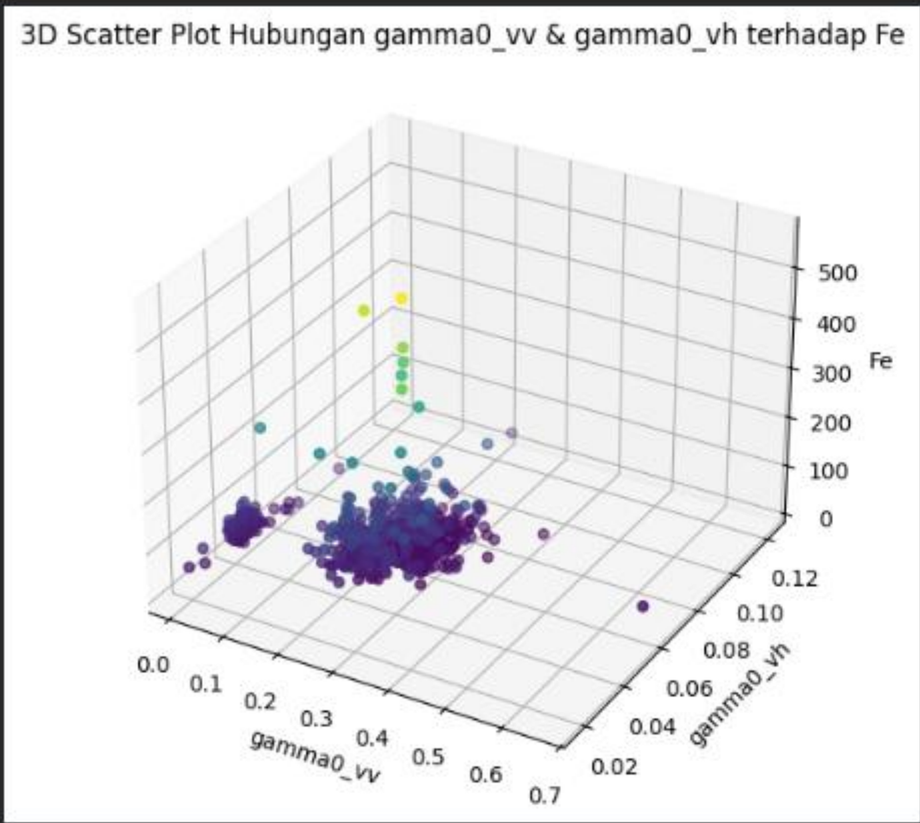
Mengukur performa model menggunakan MAE, MSE, RMSE, R^2 dan menghitung akurasi persentase.

```
plt.figure(figsize=(6,5))
plt.scatter(y_test, y_pred, edgecolor='k')
plt.xlabel("Nilai Aktual Fe")
plt.ylabel("Nilai Prediksi Fe")
plt.title("Perbandingan Nilai Aktual vs Prediksi Fe (Multiple Linear Regression)")
plt.show()
```



Grafik scatter menunjukkan seberapa dekat hasil prediksi terhadap nilai sebenarnya.

```python
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df['gamma0_vv'], df['gamma0_vh'], df['Fe'], c=df['Fe'], cmap='viridis')
ax.set_xlabel("gamma0_vv")
ax.set_ylabel("gamma0_vh")
ax.set_zlabel("Fe")
ax.set_title("3D Scatter Plot Hubungan gamma0_vv & gamma0_vh terhadap Fe")
plt.show()
```



Scatter plot 3D membantu melihat pola hubungan gamma0_vv, gamma0_vh, dan Fe.

Link GitHub: https://github.com/CeperMail/MachineLearning.git